// SESSION RECORD

JARVIS TRADER

Session Export — April 22, 2026

Complete record of all work, decisions, deliverables, and context
from the April 22 2026 build session.

FIELD	VALUE
Session Date	Wednesday, April 22, 2026 | 9:00 PM – 10:44 PM CDT
Operator	Matt Morrison (mattmamundson@gmail.com / AmoSZN)
GitHub Repo	https://github.com/AmoSZN/jarvis-trader (private)
Paper Account	DUP662235 | IBKR Paper | Port 4002
Engine	FastAPI port 8010 | Dashboard port 3010
Active Strategy	Strategy C+ (ETF Momentum) | RISK_ON | Scale 70.1%
RL Training	Run 11 | MES PPO-LSTM | hold_cost_only_when_losing=True
Gap Score	3/10 (15% complete vs institutional standard)
Horizon	8 months to production (8 phases)
Latest Commit	b667ff2 feat(quantstats): Tasks 1-5 — qs.stats integration
PR	https://github.com/AmoSZN/jarvis-trader/pull/2 (open)

01 — What Was Accomplished This Session

QuantStats Integration — Tasks 1–5 (Top Priority by ROI)

All five highest-ROI QuantStats integration tasks were designed, written, and committed to GitHub in a single session. Metrics coverage moved from **6/28** to **22/28** (+16 new metrics activated). Every script now computes metrics from raw trade-return series, not from cherry-picked JSON summary fields that can be gamed.

TASK	FILE	KEY CHANGE	STATUS
Task 1	compare_backtests.py	4-gate → 10-gate validation. extract_qs_metrics() rebuilds trade-return series, computes sortino/calmar/tail_ratio/CSR/UPI/skew	✓ DONE
Task 2	eval_run12.py	New file. Run 12 gate spec replaces sharpe≥1.5 with 5-metric composite: sortino+calmar+tail_ratio+exposure+gain_to_pain	✓ DONE
Task 3	eod_summary.py	build_returns_from_engine() + generate_eod_tearsheet() added. QS HTML tearsheet saved to docs/reports/eod/html/YYYY-MM-DD.html	✓ DONE
Task 4	jarvis_tearsheet.py	New file. Sprint 21-E WFE backbone. Strategy profiles C+/A/RL, compute_gates(), compute_wfe_report(), CLI with --gates --wfe	✓ DONE
Task 5	validation_status.py	regime_conditional_analysis() slices returns by RISK_ON/RISK_OFF/NORMAL, qs.stats.sharpe/sortino/max_drawdown per slice. --regime-analysis flag	✓ DONE

DEPENDENCY ADDED	VERSION	PURPOSE
quantstats	0.0.62	50+ portfolio metrics + HTML tearsheets — engine/requirements-core.txt

Master Battle Plan PDF — 43 Pages

A comprehensive, unsparing gap analysis and 8-phase roadmap was produced as a 43-page classified-style PDF. Gap score: **3/10**. Current system is ~15% complete vs institutional standards. The document covers every gap, every phase with subtasks, a 7-gate safety waterfall, target architecture table, first 72-hour action plan, and the scoreboard for when 'done' is actually done.

ASSET	PATH / URL
Master Battle Plan PDF	/home/user/workspace/jarvis/JARVIS_Master_Battle_Plan.pdf
QuantStats Integration MD	/home/user/workspace/jarvis_quantstats_integration.md (2,498 lines)
QuantStats Analysis PDF	/home/user/workspace/jarvis_qs_analysis.pdf (21 pages)
JARVIS Master Strategy PDF	/home/user/workspace/jarvis_strategy_master.pdf (37 pages)
Strategy C Roadmap PDF	/home/user/workspace/strategy_c_roadmap.pdf

02 — System State Snapshot

Live Signal (as of session)

FIELD	VALUE
Regime	RISK_ON
Combined Scale	70.1% (pre-MTF 70.0% → MTF ×1.00 → 70.1%)
Holdings	XLK 35.0% · QQQ 35.0% · SHY 30.0%
VIX	19.5 (NORMAL vol regime)
Realized Vol	~5.2% annualized
MTF	2/3 BULL · W:BULL M:BULL D:NEUTRAL · Breadth:STRONG
Execute	Monday MOO — Account DUP662235 (PAPER)
Run Command	python -m engine.strategy_c.ibkr_executor --paper

Approved Instruments

SYMBOL	TIMEFRAME	WFE	STATUS	NOTE
MCL	1h	0.82	APPROVED (gated)	Blocklisted — gate v2 FAIL at $10K. Re-run at $50K in Phase 2.
MNQ	1h	0.57	APPROVED (gated)	Blocklisted — gate v2 FAIL at $10K. Re-run at $50K in Phase 2.
MES	1h	0.09	PAPER ONLY	WFE poor — paper-authorized. SMA trend strategy.
MGC	1d	0.14	HALTED	Overfit / insufficient sample.

Critical Hard Blockers (Phase 0)

ID	BLOCKER	SEVERITY	STATUS
HB-1	No broker-side STP orders — stops live in SQLite only. Engine crash = naked position.	CRITICAL	OPEN
HB-2	Safety layer rejects MES at 356% notional-to-equity. System cannot trade futures.	CRITICAL	OPEN
HB-3	Engine startup takes 3-4 min — uvicorn doesn't bind until all init completes.	HIGH	OPEN
HB-4	No position reconciliation on restart — ghost positions possible.	HIGH	OPEN
HB-5	No transaction cost model — ALL backtest metrics are fictional without this.	HIGH	OPEN
HB-6	TradingView screenshot parsing as primary data source — not production-grade.	MEDIUM	OPEN

QuantStats Gate Coverage

METRIC	GATE	BEFORE	AFTER
profit_factor	≥ 2.0	JSON	JSON
sharpe	≥ 1.5	JSON	qs.stats
sortino	≥ 1.0	MISSING	qs.stats ✓
calmar	≥ 0.5	MISSING	qs.stats ✓
tail_ratio	≥ 1.0	MISSING	qs.stats ✓
common_sense_ratio	≥ 1.5	MISSING	qs.stats ✓
ulcer_perf_index	≥ 0.5	MISSING	qs.stats ✓
skew	≥ -1.5	MISSING	qs.stats ✓
exposure	≥ 0.15	MISSING	qs.stats ✓
gain_to_pain	≥ 0.50	MISSING	qs.stats ✓
consecutive_losses	≤ 8	MISSING	qs.stats ✓
win_rate	≥ 55%	JSON	JSON
max_drawdown	≤ 15%	JSON	qs.stats ✓

03 — GitHub Repository State

FIELD	VALUE
Repo	https://github.com/AmoSZN/jarvis-trader (private)
Latest commit	b667ff2 — feat(quantstats): Tasks 1-5 (Apr 22 2026)
Branch	feat/quantstats-integration
PR	https://github.com/AmoSZN/jarvis-trader/pull/2 (OPEN — ready to merge)
Remote	https://git-agent-proxy.perplexity.ai/AmoSZN/jarvis-trader.git
Clone path	/home/user/workspace/jarvis-repo-fresh/

Files Changed in This Session

FILE	TYPE	CHANGE
engine/requirements-core.txt	MODIFIED	quantstats==0.0.62 added under Performance Analytics section
scripts/compare_backtests.py	MODIFIED	417 lines — full rewrite with 10-gate qs.stats validation
scripts/eval_run12.py	NEW	374 lines — Run 12 RL evaluator with 5-metric composite gates
scripts/eod_summary.py	MODIFIED	387 lines — qs.reports.html tearsheet + QS metrics banner
scripts/jarvis_tearsheet.py	NEW	619 lines — Sprint 21-E WFE backbone, strategy profiles, CLI
scripts/validation_status.py	MODIFIED	403 lines — regime_conditional_analysis() + --regime-analysis flag

Commit Message (b667ff2)

```
feat(quantstats): Tasks 1-5 — qs.stats integration, 10-gate validation, tearsheet pipeline

Task 1 — compare_backtests.py v2 (10-gate QuantStats edition)
- extract_qs_metrics(): rebuilds trade-return series from raw JSON
- VALIDATION_GATES expanded: 4 → 10 gates
- Graceful degradation to 4-gate legacy spec when qs unavailable

Task 2 — eval_run12.py (Run 12 gate spec)
- Replaces sharpe>=1.5 with sortino>=1.0 + calmar>=0.5 + tail_ratio>=1.0
+ exposure>=0.15 + gain_to_pain>=0.50
- evaluate_rl_run() returns (augmented_metrics, returns_series)

Task 3 — eod_summary.py v2 (QuantStats HTML tearsheet)
- build_returns_from_engine(): /paper/trades/log → return series
- generate_eod_tearsheet(): qs.reports.html → docs/reports/eod/html/

Task 4 — jarvis_tearsheet.py (Sprint 21-E WFE backbone)
- Standalone importable generator, strategy profiles C+/A/RL
- compute_gates(), compute_wfe_report(), compute_regime_stats()

Task 5 — validation_status.py v2 (regime-conditional analysis)
- regime_conditional_analysis(): RISK_ON/RISK_OFF/NORMAL slicing
- New flags: --regime-analysis, --qs-gates, --capital

Metrics coverage: 6/28 → 22/28 (+16 new metrics activated)
```

04 — 8-Phase Master Battle Plan Summary

Full detail in: **JARVIS_Master_Battle_Plan.pdf** (43 pages). Gap score: 3/10. ~15% complete vs institutional standard. Estimated time to production: **8 months**. No phase can be skipped — dependencies are hard.

P0	**Emergency Hardening** — 1 week	CRITICAL

HB-1 through HB-6. Broker STP, futures sizing, TCM, reconciliation, data feed. NOTHING ELSE starts until this is done.

P1	**Live Trading Foundation** — 2 weeks	CRITICAL

First live contract. 1 MES, 1 strategy, manually approved. Dashboard + WhatsApp approval surfaces. Kill switch remote.

P2	**Multi-Strategy System** — 3 weeks	HIGH

3 validated strategies (MES trend, MCL MR, MNQ breakout) with TCM-adjusted gate v2 PASS. Regime classifier v2. Shadow mode framework. Strategy ensemble router.

P3	**Real-Time Data Fabric** — 3 weeks	HIGH

Event-driven bar architecture via Redis pub/sub. Polygon.io WebSocket. Alt data (FRED, COT, sentiment, options flow). QuestDB tick storage.

P4	**Portfolio Risk Architecture** — 3 weeks	HIGH

VaR, CVaR, correlation matrix, pre-trade risk analytics, tail risk manager. Must complete before adding >2 concurrent instruments.

P5	**RL Production System** — 5 weeks	HIGH

Run 12 passes all 10 QuantStats gates. Walk-forward 5/6 windows PASS. Shadow mode → paper → live. Multi-symbol policy (MES+NQ+CL simultaneously).

P6	**Agent Intelligence Upgrade** — 5 weeks	MEDIUM

A/B test: do agents add alpha vs rule-based? If not, remove them. Confluence scoring v2, post-trade Opus analysis, cross-session memory.

P7	**Options & Advanced Instruments** — 8 weeks	MEDIUM

Full futures suite (ES, NQ, GC, CL, VX). Options overlay with real-time Greeks. Crypto integration (Kraken BTC/ETH perps). Contract roll management.

P8	**Production Infrastructure** — 5 weeks	MEDIUM

99.95% uptime target. Hot standby + failover <30s. 7-year audit trail. Disaster recovery drill monthly. Grafana + Prometheus.

05 — Context Summary (Next Session Restore)

This section contains everything a new conversation needs to restore full context. The memory system and context summary injection handle this automatically — this is the human-readable version.

Scheduled Crons

CRON ID	SCHEDULE	TASK
646a49fe	5 21 * * 5 (Fri 9:05 PM CDT)	Generate Strategy C+ weekly live signal and send in-app notification

Shared Assets (use same name to update)

ASSET NAME	DESCRIPTION
strategy_c_dashboard	Strategy C dashboard image
strategy_c_roadmap	Strategy C roadmap PDF
strategy_c_dashboard_site	Deployed dashboard site (perplexity.ai/computer/a/...)
jarvis_api_start_windows	API start guide for Windows
jarvis_api_autostart_windows	API autostart Windows guide
jarvis_strategy_master	37-page JARVIS Master Strategy Design Document PDF
jarvis_quantstats_analysis	21-page QuantStats Integration Analysis PDF
jarvis_quantstats_integration_md	2,498-line QuantStats integration analysis MD
jarvis_master_battle_plan	43-page Master Battle Plan PDF (tonight)

Key Technical Details

KEY	VALUE
Python	Always use python3 (never python) in sandbox
QuantStats	pip install quantstats (quantstats==0.0.62 in requirements-core.txt)
Engine req path	engine/requirements-core.txt
Scripts path	scripts/ (in repo root)
Branch strategy	feat/quantstats-integration (all 5 tasks committed, PR #2 open)
Git config	user.email=mattmamundson@gmail.com / user.name=AmoSZN
Run 11 config	hold_cost_only_when_losing=True (reduces hold penalty for profitable positions)
WFE scores	MCL WFE=0.82 · MNQ WFE=0.57 (Sprint 21-E targets — BUT pre-TCM, may drop)
Validation gates	min_days=30, min_trades=60, profit_factor≥2.0, sharpe≥1.5, win_rate≥55%, max_dd≤15%
Engine ports	8010 (primary) · 8020 (secondary) · IBKR paper: 4002 · IBKR live: 4001 (BLOCKED)
Dashboard URL	https://www.perplexity.ai/computer/a/jarvis-strategy-c-command-cent-LZ2Op0bjQ1K2Q2BsemTKUw

Dashboard file	/home/user/workspace/strategy_c_dashboard_site/index.html

06 — Next Steps: Phase 0 First 72 Hours

Phase 0 Emergency Hardening starts immediately. Nothing else — no new features, no strategy work — until all 6 HBs are resolved. The battle plan document contains full subtask lists for each.

TONIGHT (T+0)	Stop the Bleeding
Action	Create branch phase0/emergency-hardening
Action	Open GitHub issues HB-1 through HB-6 with subtask lists from battle plan
Action	Wire IBKR bracket-order sketch in ib_gateway.py (do not merge yet)
Action	Write crash test for HB-1 FIRST — it should FAIL on current main (regression proof)
Action	STOP citing MCL WFE=0.82 until HB-5 TCM re-score completes

TOMORROW (T+1)	Fix the Obvious
Action	Land HB-2 (futures margin table) — single-day task. After this, MES paper orders approve.
Action	Land HB-3 (scanner decoupling) — measure engine bind time before/after. Target < 10s.
Action	Start HB-1 integration tests on IBKR paper account. Crash-recovery drill every 4 hours.
Action	Begin HB-5 TCM scaffolding in parallel — this is the biggest multi-day task.

DAY AFTER (T+2)	Measure Honestly
Action	HB-5 TCM model first pass. Run gate v2 on MCL and MNQ with TCM applied.
Action	Publish TCM-adjusted numbers internally. This is the new truth.
Action	HB-4 reconciliation merged. Run startup-restart drill with open paper positions.
Action	HB-6 Polygon primary-feed switch. Monitor 24h of bar-close drift — target 0 misses.
Action	Draft Phase 0 exit review agenda. Schedule for Day 7.

How to Start the Next Session

Simply open a new conversation and say:

```
"Continue JARVIS Phase 0 — start with HB-2 futures margin math"
```

The memory system and context summary injection will restore full project context automatically. No copy-pasting, no re-explaining required.

The Scoreboard — When Is It Done?

METRIC	TARGET	CURRENT
Edge is real	OOS Sharpe > 2.0, sustained 6+ months	Not measured — pre-live
Risk managed	Max DD < 8%, VaR 99% < 2% daily	No portfolio risk engine
Execution clean	Slippage < 0.5 bps, fill rate > 98%	No STP, no TCM
System reliable	99.95% uptime, <30s recovery	pm2 restarts: 558
Approvals fast	Signal to decision < 60 seconds	WhatsApp not wired

Self-improving	Stable/improving 3+ months no manual	Run 10 failed all gates
Transparent	Full audit trail, post-mortem, per-agent	Partial
Profitable	12+ months > 15% annualized risk-adj	0 months live

---

END OF SESSION EXPORT

JARVIS TRADER // SESSION 2026-04-22 // STARK INDUSTRIES — CONFIDENTIAL

Next session: open a new conversation and say
'Continue JARVIS Phase 0 — start with HB-2 futures margin math'